

Subregional Analysis of Great Lakes Ice Cover: Connecting Ice to Local Weather Data

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Introduction

Ice cover in the Great Lakes highly variable and has significant influence on regional climate, ecology, and economy. Spatially resolved ice cover ($\Delta x = 1.25\text{KM}^2$) has been recorded since 1973 and is available as daily files throughout each ice season¹. Previous work has assessed large-scale (i.e. lake or basin-wide) ice cover fluctuations through time and their relationship to global weather phenomena^{2,3,4,5,6}. Following Mason et al. (2016)² we use the Great Lakes Aquatic Habitat Framework (GLAHF)⁷ to analyze ice cover and local meteorological data for 1973-present and ask two research questions:

1. Do significant ice trends exist at these spatial scales?
2. Are correlations between atmosphere and ice data consistent with our understanding of physical processes?

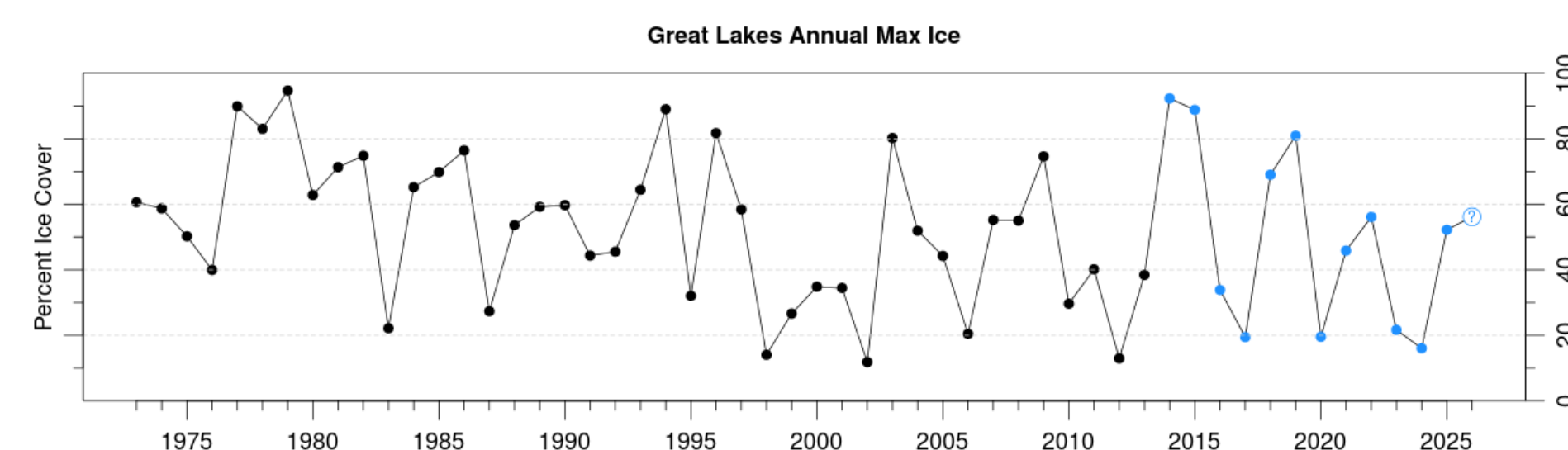


Figure 1. Annual Max Ice cover for the Great Lakes (ice years after 2013 highlighted in blue).

Methods

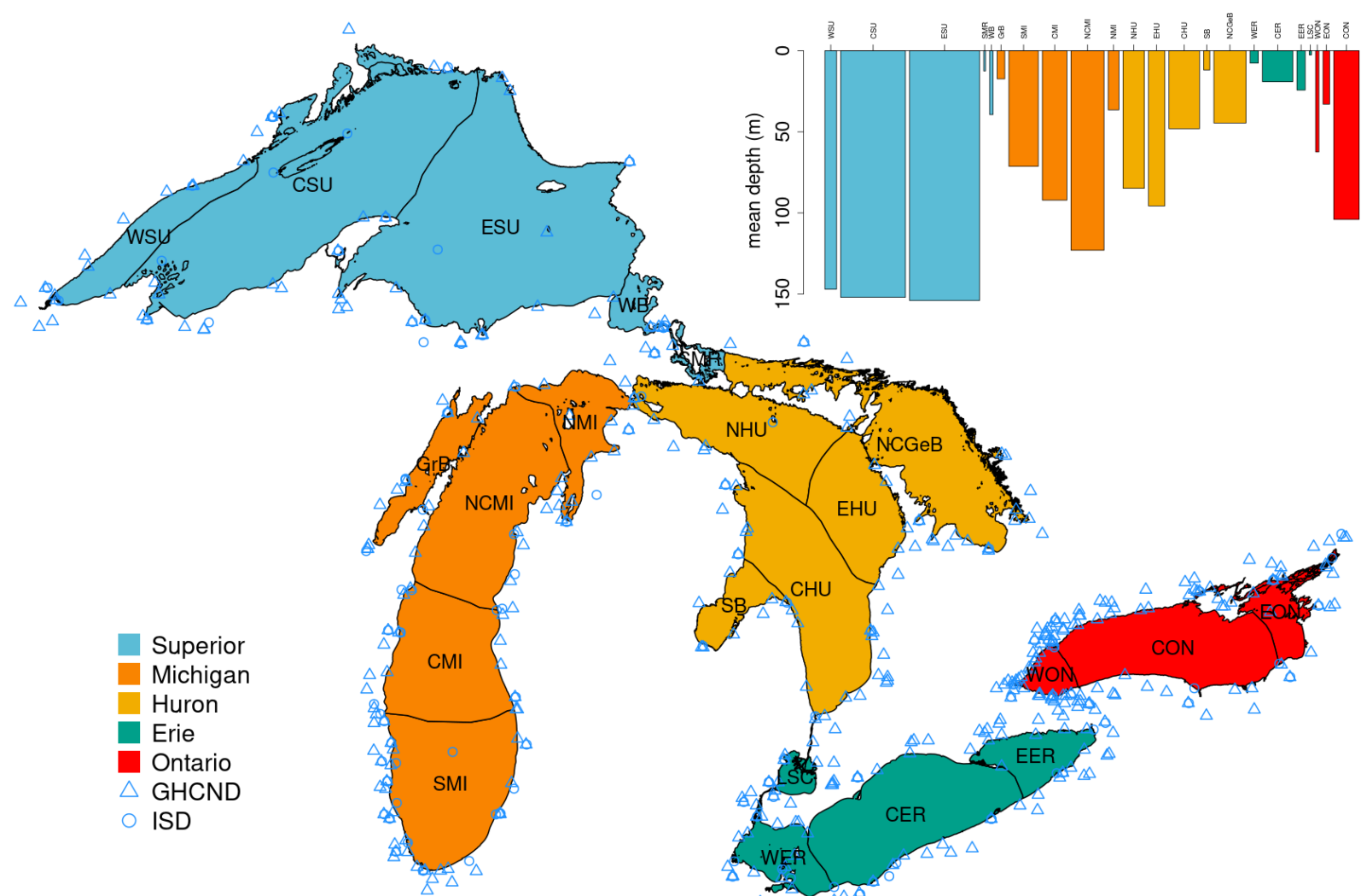


Figure 2. Spatial delineation of GLAHF subregions over water in relation to selected met station locations from the Global Historical Climatology Network Database⁸ and the Integrated Surface Database⁹. The upper right figure shows depth and (relative) surface area of each subregion in

To address the first research question, the below metrics are calculated for each GLAHF subregion and ice season and then Mann-Kendall monotonic trend test¹⁰ is applied to check for statistical significance:

- **Annual Max Ice Cover (AMIC):** maximum percent during a given season
- **Seasonal Average Ice Cover (JFM):** mean daily ice cover: Jan 1 - Mar 31
- **Ice Duration:** Total (non-consecutive) days[†] where ice cover exceeded 10%

To address the second research question, station data^{8,9} is aggregated for each subregion based on a 20 KM radius and correlations between ice metrics and daily mean meteorological surface variables are identified. On average, 30 stations per subregion were located within the radius but station data availability varied through time, location and variable. Temperature min/max were reported most frequently (median station count ≈ 10 per subregion). Other variables—cloud cover, precipitation, wind speed, were reported less frequently (median ≈ 5).

[†]: days are only counted Jan 1 - Mar 31 due to shorter ice reporting windows during 1970's and 80's. This method likely *underestimates* trends in decreasing ice duration but is required to avoid artificial season length decrease.

Analysis

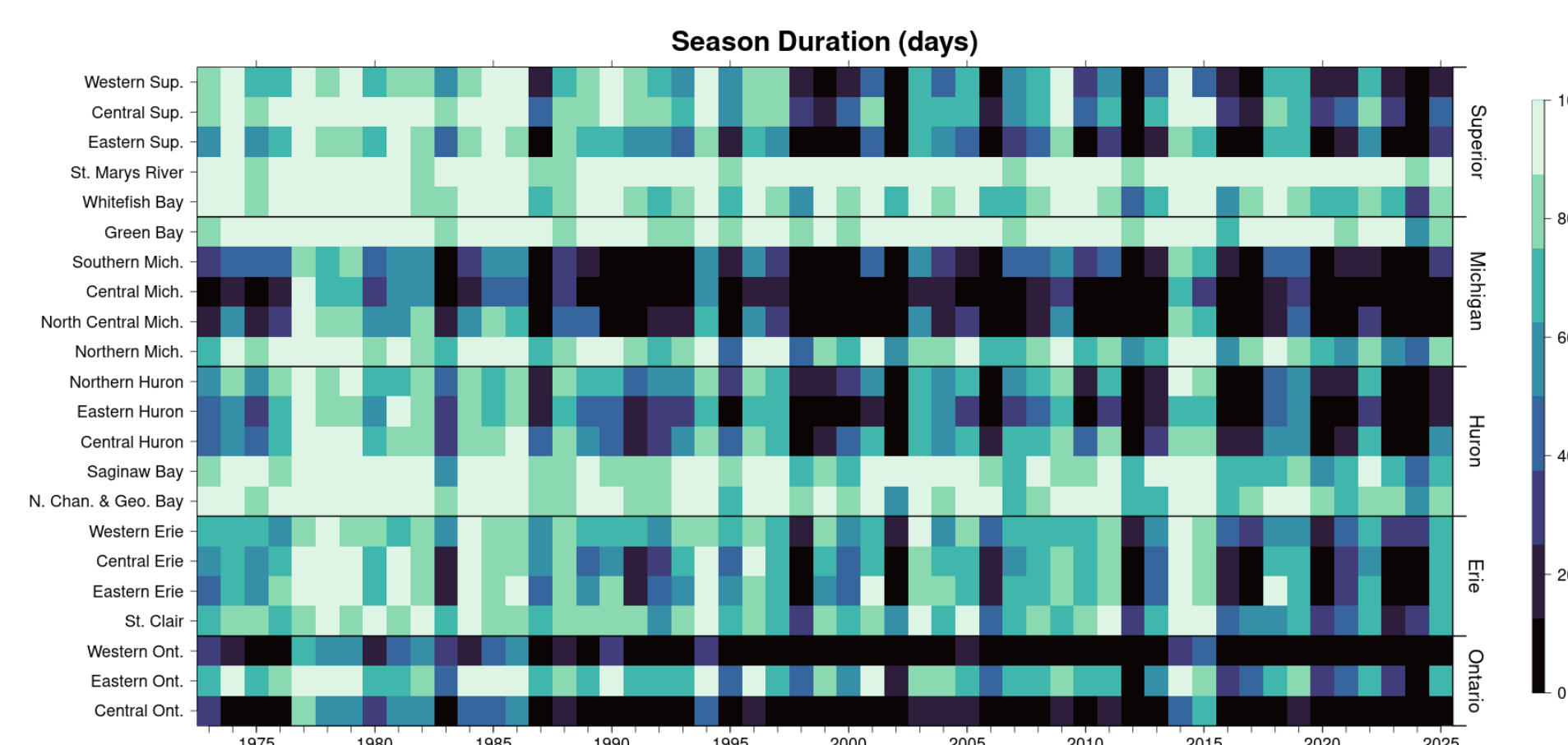


Figure 3. Ice duration for each subregion.

Season duration decrease is visible in some regions but not apparent in others (see fig. 3). Regions that show decrease are often offshore and deep: Superior (WSU, CSU, ESU), Huron (NHU, EHU, CHU), and Ontario (WON, CON). This may be due to warmer subsurface temps^{11,12} during later years providing heat to surface waters and preventing freezing, whereas shallow regions may be able to consistently release all heat from the water column.

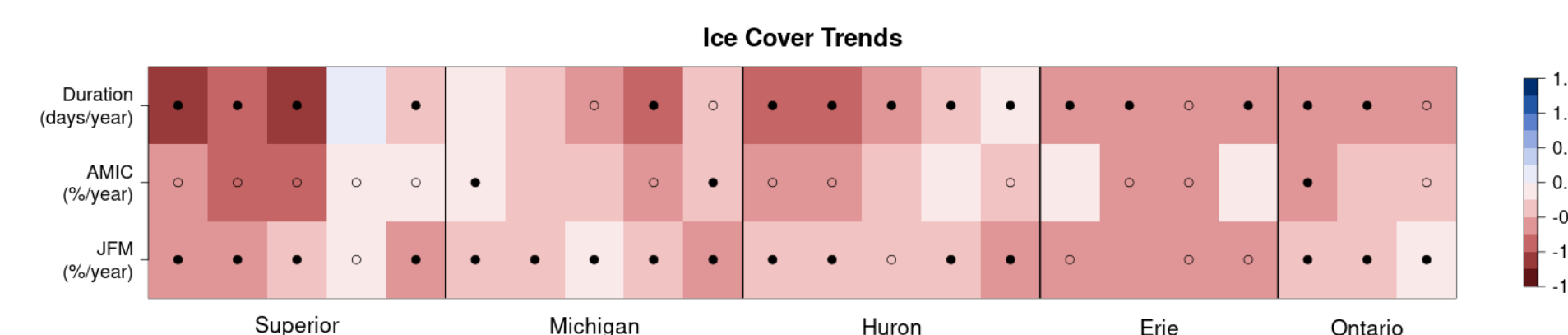


Figure 4. Linear trends for each metric. Open (solid) dots indicate 95% (99%) statistical significance. Columns correspond to rows in fig 3.

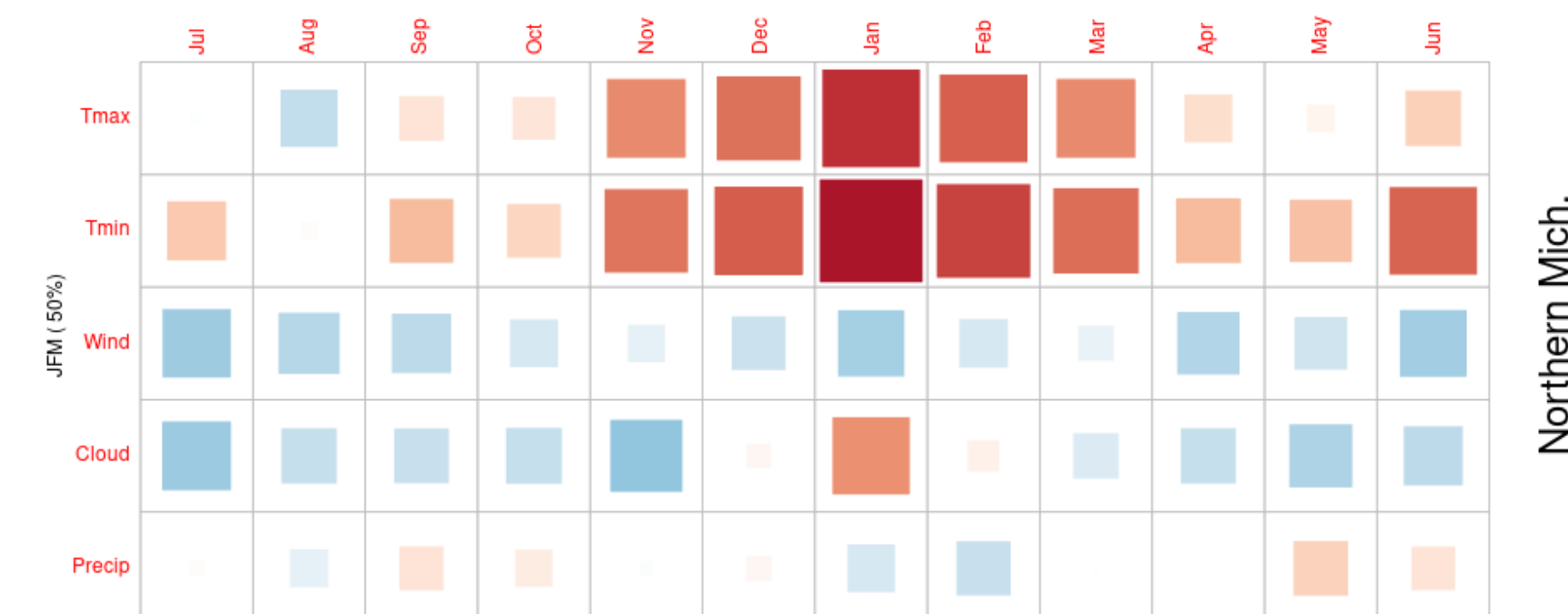


Figure 5. Correlation between Seasonal Ice Cover (JFM) and local meteorology. Meteorological variables are aggregated by month (columns) and state variable (rows). Larger squares indicate stronger correlations and blue (red) indicates direct (indirect) correlation.

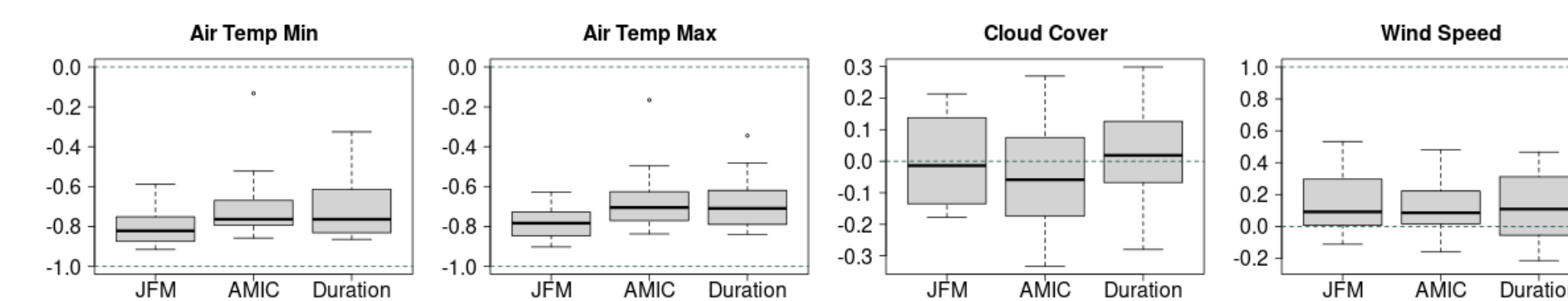


Figure 6. Distribution (N=22 subregions) of correlations (Pearson's r) between three ice metrics and met station variables.

As expected, ice cover showed strong inverse correlation with air temperature. Wind speed showed moderate correlation to ice in some regions which may be due to wind's influence on heat distribution in the water column via mixing (though this impact is more relevant during fall overturn). The analyses presented here are a small fraction of possible correlation pairs and more work is needed, including different time periods and other metrics like ice onset/offset.

Summary

Trend assessment:

- A majority of subregions show significant decreasing trends in all three metrics
- Seasonal average trends were significant in ALL 22 subregions
- Max Ice Cover showed the least significant trends (17/22 regions)

Met Data Correlation:

- Air temperature was strongly anti-correlated with ice cover for all metrics
- More work needed: comprehensive assessment of ice cover and met data correlations

References

- [1] Ting-Yi Yang et al. "A consistent Great Lakes ice cover digital data set for winters 1973–2019". In: *Scientific Data* 7.1 (Aug. 2020). DOI: 10.1038/s41597-020-00603-1.
- [2] Lacey A. Mason et al. "Fine-scale spatial variation in ice cover and surface temperature trends across the surface of the Laurentian Great Lakes". In: *Climatic Change* 138.1–2 (June 2016), pp. 71–83. DOI: 10.1007/s10584-016-1721-2.
- [3] Sapna Sharma et al. "Widespread loss of lake ice around the Northern Hemisphere in a warming world". In: *Nature Climate Change* 9.3 (Jan. 2019), pp. 227–231. DOI: 10.1038/s41558-018-0393-5.
- [4] Catherine M. O'Reilly et al. "Rapid and highly variable warming of lake surface waters around the globe". In: *Geophysical Research Letters* 42.24 (Dec. 2015). DOI: 10.1002/2015GL066295.